

Jaehyun Yoon

Mechanical Engineering | Robotics Researcher

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Research Interests

Continuum Robotics • Medical Robotics • Bio-Inspired Robotics • Compliant Mechanisms • Smart Material Actuation • Haptic Interfaces

Education

Korea University

M.S. in Mechanical Engineering

2026 – Present

Seoul, Republic of Korea

Konkuk University

B.S. in Mechanical Engineering

2019 – 2026

Seoul, Republic of Korea

Research Experience

Korea University

Graduate Researcher, Department of Mechanical Engineering

2026 – Present

Seoul, Republic of Korea

- Developing a **bio-inspired modular continuum robot** with independently controllable mechanical locking mechanisms for localized shape reconfiguration and stable posture maintenance.
- Designed an **SMA-actuated locking/unlocking mechanism** combined with tendon-driven actuation to selectively rotate individual robot segments.
- Investigating mechanically locked robotic structures capable of maintaining loaded configurations without continuous actuator input.
- Exploring applications of reconfigurable robotic mechanisms in **medical robotics, continuum manipulation, and adaptive robotic systems**.

Publications

- J. H. Yoon**, J. S. Bae, J. H. Choi, S. Kang, J. Lee, and T. H. Yang, "Development of a Portable Commerce Photography Automation Robot," *Journal of the Korean Society for Precision Engineering*, vol. 43, no. 7, pp. 689–700, 2026. [\[DOI\]](#)
- "MORPH: MR-Fluid Operated Reconfigurable Programmable Haptic Grip Interface for Bidirectional and Multimodal Grasping Interaction in Telerobotics and XR," *Advanced Intelligent Systems*, **Under Review**.
- "A Parametrically Scalable Compliant Gripper for Fruit Harvesting: Mold-Free Additive Manufacturing with Reduced Material and Process Waste," *International Journal of Precision Engineering and Manufacturing-Green Technology*, **Submitted**.

Selected Research Projects

Bio-Inspired Modular Continuum Robot

Medical Robotics / Continuum Robotics

- Developed a multi-segment robotic architecture inspired by the regional motion of a monkey tail.
- Integrated tendon-driven actuation with independently controlled mechanical locks to enable selective segment rotation.
- Designed an SMA-based unlocking mechanism that allows individual segments to transition between movable and mechanically locked states.
- Investigated workspace expansion, posture stabilization, obstacle avoidance, and surgical-camera applications.

MORPH: MR-Fluid Haptic Grip Interface

Haptics / Telerobotics / XR

- Developed a reconfigurable and programmable haptic grip interface based on **magnetorheological (MR) fluid**.
- Designed the interface for bidirectional and multimodal grasping interaction in telerobotic and extended-reality applications.
- Investigated programmable physical interaction through controllable mechanical properties of MR-fluid-based structures.

Portable Commerce Photography Automation Robot

Robotic Automation / E-Commerce

- Developed a portable **4-DOF robotic photography system** for standardized e-commerce product imaging.
- Automated camera height and tilt, object rotation, and perspective translation for consistent multi-view image acquisition.
- Implemented automatic camera-pose planning according to object dimensions.
- Demonstrated repeatable image acquisition with pairwise Intersection over Union (IoU) values above **0.90**.

Parametrically Scalable Compliant Gripper

Compliant Mechanisms / Additive Manufacturing

- Developed a compliant robotic gripper with a parametrically scalable structure for fruit-harvesting applications.
- Investigated mold-free additive manufacturing to reduce fabrication complexity, material waste, and process waste.

Hyundai AutoEver Smart Mobility Mentoring Program

Smart Mobility

- Participated in a smart-mobility engineering program focused on practical engineering problem solving and mobility-system development.

Team KAI Autonomous Mobility Project

Autonomous Mobility

- Participated in the development and competition activities of an autonomous mobility platform as a member of **Team KAI**.

Awards & Honors

Runner-Up (2nd Place), 2026 Korea University K-3MT Competition

2nd place among 45 graduate student participants

2026

Korea University

- Selected as Runner-Up for effectively communicating graduate research to a non-specialist audience in a three-minute presentation.

Junior Fellowship Research Grant (JFR)

Research Grant Recipient, Graduate School

2026

Korea University

Technical Skills

CAD & Mechanical Design	Fusion 360, Autodesk Inventor, CATIA
Rapid Prototyping	FDM, SLA, Additive Manufacturing, Mechanical Prototyping
Robotics	Continuum Robots, Tendon-Driven Mechanisms, Compliant Mechanisms, Mechanical Locking Mechanisms
Smart Materials	Shape Memory Alloy (SMA), Magnetorheological (MR) Fluid
Experimental Systems	Laser Displacement Sensing, Thermal Imaging, Current/Voltage Measurement, Robotic Experimentation
Research Areas	Medical Robotics, Bio-Inspired Robotics, Haptics, Smart-Material Actuation